AOU
الجامعة العربية المفتوحة
Arab Open University

Faculty of Business Studies

BUS102

Introduction to Statistics

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MTA Examination - Makeup / Version: A

2024-2025 / Spring

22-April-2025

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed: 1.5 hours**

Instructions:

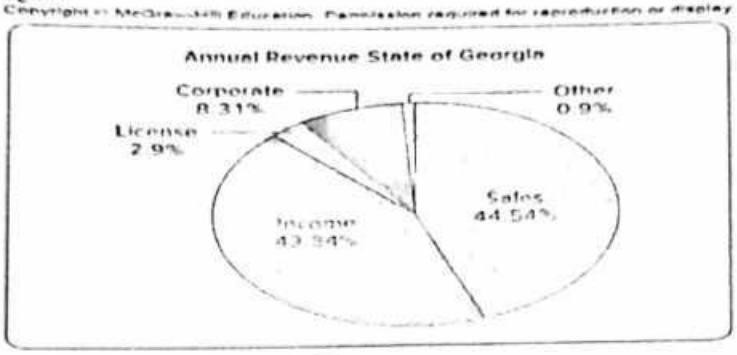
1- Take **FIVE (5)** minutes to read all instructions carefully before attempting to write.
2- Solve all FOUR given questions.
3- Mobile phones must be in "SWITCHED **OFF**" mode throughout the examination.
4- This is a **CLOSED** book examination. NO documents of any kind are allowed.
5- You **MUST** answer the questions using only the answer booklets provided
6- You **MUST** return the examination paper together with your answer booklets.
7- Do **NOT** write your answers in pencil except graphs/sketches
8- Students are **NOT** allowed to use any type of dictionary or mobile phones during the examinatio
9- Students are allowed to use ***non-programmable*** calculators only (if needed).
10- Any form of cheating will **NOT** be tolerated and will subject you to AOU Cheating Policy.

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Question 1 (30 marks)

A. The following chart depicts the annual revenues, by type of tax, for the state of Georgia.

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(15 marks)

a. What is the chart called?
b. What percentage of the state revenue is made by sales tax and income tax?
c. Which category will generate more revenue: corporate taxes or license fees?
d. The total annual revenue for the state of Georgia is $6.3 billion. Estimate the amount of revenue in billions of dollars for sales taxes and for income taxes.

A. The frequency distribution of order lead time of a retailer is repeated below

Lead Time (days)	Frequency
0 up to 5	6
5 up to 10	7
10 up to 15	12
15 up to 20	8
20 up to 25	7
Total	40

(15 marks)

a. How many orders were filled in less than 15 days?
b. Convert the frequency distribution to cumulative frequency and cumulative relative frequency distributions.
c. Develop a cumulative frequency polygon.
d. About 60% of the orders were filled in less than how many days?

Question 2 (40 marks)

A. A sample of households that subscribe to United Bell Phone Company for landline phone service revealed the following number of calls received per household last week.

28 39 23 67 37 28 56 (25 marks)

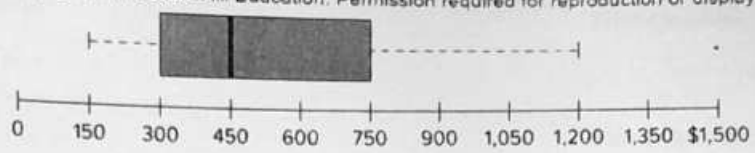
a. Explain The difference between the sample and the population.
b. Find the mean and median of the number of calls.
c. Find the Variance and the standard deviation of the number of calls.
d. Determine the first and third quartiles of the number of calls.
e. Find the range and determine the interquartile range of the number of calls.
f. Compute the skewness value and judge the skewness of the distribution.
g. Find the coefficient of variation of the number of calls.

B. The box plot shows the undergraduate in-state tuition per credit hour at four-year public colleges.

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(15 marks)

a. Estimate the median.
b. Estimate the first and third quartiles.
c. Determine the interquartile range.
d. Beyond what point is a value considered an outlier?
e. Is the distribution symmetrical or positively or negatively skewed?

Question 3 (30 marks)

A. A survey of 545 college students asked: What is your favorite winter sport? And, what type of college do you attend? The results are summarized below:

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	Favorite Winter Sport			
College Type	Snowboarding	Skiing	Ice Skating	Total
Junior College	68	41	46	155
Four-Year College	84	56	70	210
Graduate School	59	74	47	180
Total	211	171	163	545

(15 marks)

Using these 545 students as the sample, a student from this study is randomly selected.

a. If the student selected is a four-year-college student, what is the probability that the student prefers ice skating?
b. If the student selected prefers snowboarding, what is the probability that the student is in junior college?
c. If a graduate student is selected, what is the probability that the student prefers skiing or ice skating?

B. All Seasons Plumbing has two service trucks that frequently need repair. If the probability the first truck is available is 0.75, the probability the second truck is available is 0.50, and the probability that both trucks are available is 0.30.

(15 marks)

a. what is the probability neither truck is available?
b. what is the probability that only one truck is available?
c. what is the probability that both of the trucks will be available?

END OF EXAM QUESTION

FORMULAS

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Sample Mean

$$\overline{X} = \frac{\sum_{i=1}^{n} X_i}{n} \quad (3.1)$$

Median

$$\text{Median} = \frac{n+1}{2} \text{ ranked value} \quad (3.2)$$

Geometric Mean

$$\overline{X}_G = (X_1 \times X_2 \times \cdots \times X_n)^{1/n} \quad (3.3)$$

Geometric Mean Rate of Return

$$\overline{R}_G = [(1 + R_1) \times (1 + R_2) \times \cdots \times (1 + R_n)]^{1/n} - 1 \quad (3.4)$$

Range

$$\text{Range} = X_{\text{largest}} - X_{\text{smallest}} \quad (3.5)$$

Sample Variance

$$S^2 = \frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{n-1} \quad (3.6)$$

Sample Standard Deviation

$$S = \sqrt{S^2} = \sqrt{\frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{n-1}} \quad (3.7)$$

Coefficient of Variation

$$CV = \left(\frac{S}{\overline{X}}\right)100\% \quad (3.8)$$

Z Score

$$Z = \frac{X - \overline{X}}{S} \quad (3.9)$$

First Quartile, Q_1

$$Q_1 = \frac{n+1}{4} \text{ ranked value}$$

Third Quartile, Q_3

$$Q_3 = \frac{3(n+1)}{4} \text{ ranked value}$$

Interquartile Range

$$\text{Interquartile range} = Q_3 - Q_1$$

Population Mean

$$\mu = \frac{\sum_{i=1}^{N} X_i}{N}$$

Population Variance

$$\sigma^2 = \frac{\sum_{i=1}^{N} (X_i - \mu)^2}{N}$$

Population Standard Deviation

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N} (X_i - \mu)^2}{N}}$$

Sample Covariance

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^{n} (X_i - \overline{X})(Y_i - \overline{Y})}{n-1}$$

Sample Coefficient of Correlation

$$r = \frac{\text{cov}(X, Y)}{S_X S_Y}$$

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Probability of Occurrence

$$\text{Probability of occurrence} = \frac{X}{T} \quad (4.1)$$

Marginal Probability

$$P(A) = P(A \text{ and } B_1) + P(A \text{ and } B_2) + \cdots + P(A \text{ and } B_k) \quad (4.2)$$

General Addition Rule

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B) \quad (4.3)$$

Conditional Probability

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)} \quad (4.4a)$$

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)} \quad (4.4b)$$

Independence

$$P(A|B) = P(A) \quad (4.5)$$

General Multiplication Rule

$$P(A \text{ and } B) = P(A|B)P(B) \quad (4.6)$$

Multiplication Rule for Independent Events

$$P(A \text{ and } B) = P(A)P(B) \quad (4.7)$$

Marginal Probability Using the General Multiplication Rule

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \cdots + P(A|B_k)P(B_k)$$

Bayes' Theorem

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \cdots +}$$

Counting Rule 1

$$k^n$$

Counting Rule 2

$$(k_1)(k_2)\ldots(k_n)$$

Counting Rule 3

$$n! = (n)(n-1)\ldots(1)$$

Counting Rule 4: Permutations

$${}_nP_x = \frac{n!}{(n-x)!}$$

Counting Rule 5: Combinations

$${}_nC_x = \frac{n!}{x!(n-x)!}$$

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